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ABSTRACT

The adoption of Electric Vehicles (EVs) has emerged quickly and therefore, prompted the demand for enhanced battery safety features and fault prevention mechanisms. Battery malfunction - Overcharging, deep discharge, overheating and fire risks are all serious hazards both to vehicles and passengers. This paper demonstrates the Voice Alert System for Fire and Fault Prevention in battery operated EVs using Internet of Things (IoT) technologies to carry out real-time monitoring and automated safety interventions. The proposed system consists of the integration of voltage, temperature and flame sensors with a NodeMCU microcontroller, Arduino, relay module, LCD display, and a buzzer that provide continuous monitoring of several parameters of the battery like voltage, current, temperature, state of charge (SoC), and state of health (SoH). When abnormal conditions are detected by the system such as high temperature, fire etc. Relays module directly disconnects the battery and the voice alert and buzzer sound are instantly generated to prevent escalation. The NodeMCU makes it possible to transmit data to the cloud, to monitor battery status remotely using web or mobile interfaces. Additionally, the system also logs historical data, which enables the distinction of predictive maintenance and early detection of potential failures. By using a combination of local alerts and IoT-enabled systems for remote monitoring and data analytics on the vehicle, the system ensures higher battery safety and longer battery life, while also reducing the risk of accidents. The proposed approach is scalable, cost-effective and can be integrated with different kinds of EVs to ensure better-safety and reliable-electric mobility.

Keywords: Electric Vehicle (EV) ,Battery Monitor System, IoT -Internet of Things, Fire detection System, Fault Prevention System, Real-time alert system, Predicative maintenance

1.INTRODUCTION

Electric vehicles (EVs) are transforming transportation by reducing greenhouse gas emissions and dependence on fossil fuels, but their safety and efficiency hinge on battery performance. Lithium-ion batteries, widely used in EVs, are sensitive to temperature, voltage, and environmental factors; uncontrolled conditions (overcharging, deep discharging, or overheating) can degrade capacity, shorten lifespan, or cause thermal runaway and fires. To address these risks, an IoT-based Battery Monitoring System provides real-time data acquisition and automated safety actions.

The system combines voltage, temperature, and flame sensors with microcontrollers (NodeMCU for Wi-Fi and Arduino), an LCD for local feedback, a relay for automatic disconnection, and a buzzer for alarms. Voltage monitoring prevents overcharge and deep discharge; temperature sensing detects abnormal heat that can lead to thermal runaway; flame detection identifies combustion. On detecting anomalies the system sounds alarms and disconnects the battery via the relay, while the NodeMCU transmits real-time data and alerts to a cloud server for remote monitoring via web/mobile interfaces.

Continuous logging enables predictive maintenance and trend analysis, reducing downtime and costs and extending battery life. The solution reduces human error, boosts consumer confidence in EV safety, and is adaptable to different vehicle and battery configurations. By enabling remote, data-driven battery management, the IoT monitoring system supports safer, more reliable, and more scalable electric-vehicle deployment within smart transportation ecosystems.

1.1 INTERNET OF THINGS

The Internet of Things (IoT) is a network of goods that include electronics, software, actuators, and connectivity, allowing these devices to connect, interact, and exchange data. The Internet of Things means extending Internet connectivity beyond standard devices such as PCs, laptops, smartphones, and tablets to a wide range of previously dumb or non-internet-enabled physical gadgets and everyday objects. These technologically implanted items can connect and interact over the Internet, and they can be remotely monitored and controlled.

The definition of the Internet of Things has changed as a result of the convergence of several technologies, including real-time analytics, machine learning, inexpensive sensors, and

embedded systems. The Internet of things is made possible by embedded systems, wireless sensor networks, control systems, automation (including home and building automation), and other conventional disciplines.

1.1.1 WORKING PROCESS

An Internet of Things (IoT) ecosystem is made up of web-enabled smart devices that employ embedded computers, sensors, and communication gear to gather, communicate, and act on environmental data. By connecting to an IoT gateway or other edge device, which either sends data to the cloud for analysis or analyzes it locally, IoT devices exchange the sensor data they gather. These gadgets converse with other similar devices on occasion, acting on the data they exchange. Although individuals may engage with the devices to set them up, give them instructions, or retrieve the data, the gadgets accomplish the majority of the job without their help. These web-enabled devices' connection, networking, and communication protocols are primarily determined by the particular IoT.

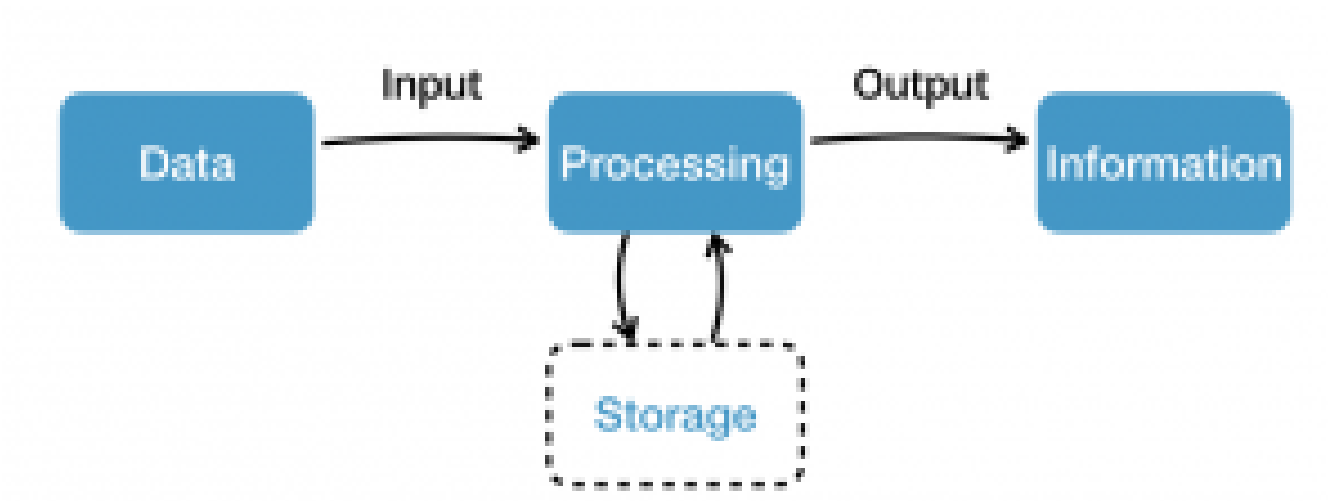

Figure 1.1-Example of an IoT system

1.1.2 ADVANTAGES

Organizations can use the internet of things to:

• monitor their overall business processes;

• improve the customer experience;

• save time and money;

• increase employee productivity;

• integrate and adapt business models;

• make better business decisions; and

• generate more revenue.

IoT pushes organizations to reconsider how they approach their enterprises, sectors, and markets, and it provides them with the means to better their business plans.

1.1.3 ISSUES

The internet of things involves the usage of billions of data points and the connection of billions of devices to the internet, all of which must be protected. IoT security and privacy are important concerns due to its extended attack surface. Mirai, a botnet that hacked domain name server provider Dyn and brought down many websites for a long period of time in one of the largest distributed denial-of-service (DDoS) assaults ever witnessed, was one of the most prominent recent IoT attacks. Attackers acquired network access by exploiting inadequately protected IoT devices.Because IoT devices are so tightly linked, a hacker only has to exploit one weakness to corrupt all of the data, rendering it useless. Manufacturers that fail to update their gadgets on a regular basis, if at all, leave them open to hackers.Furthermore, linked gadgets frequently prompt users to provide personal information such as names, ages, residences, phone numbers, and even social network accounts – information that hackers might exploit.However, hackers aren't the only threat to the internet of things; IoT consumers are also concerned about their privacy. Companies that manufacture and distribute consumer IoT devices, for example, might utilize those devices to collect and sell personal data from customers.Aside from exposing personal data, IoT threatens key infrastructure such as power, transportation, and financial services.

1.1.4 APPLICATIONS

The applications of the Internet of Things (IoT) are vast and diverse, as the technology can be applied to various industries and use cases. Some common applications of IoT include:

Smart homes: IoT devices can be used to automate and control various home appliances and systems, such as lighting, heating, and security systems.

Industrial automation: IoT can be used to optimize and automate industrial processes, such as manufacturing, logistics, and supply chain management.

Healthcare: IoT devices can be used to monitor patient health and vital signs, automate medication management, and improve patient outcomes.

Agriculture: IoT can be used to monitor soil moisture, temperature, and other environmental factors to optimize crop yields and reduce water usage.

Smart cities: IoT can be used to optimize urban infrastructure, such as traffic management, waste management, and energy usage.

Transportation: IoT can be used to optimize transportation systems, such as monitoring traffic patterns and optimizing vehicle routes.

2.EXISTING AND PROPOSED SYSTEM

2.1EXISTING SYSTEM

Traditional Battery Management Systems (BMS) in electric vehicles (EVs) monitor charging, discharging, voltage, temperature, and current to maintain battery health and prevent hazards such as overcharging, overheating, and short circuits. They use protection circuits and basic sensors to keep operations within safe limits, focusing mainly on state of charge (SOC) and state of health (SOH).

While effective for basic safety, conventional systems have key limitations:

- **Lack of real-time monitoring and remote access**: Data is processed locally, preventing predictive maintenance or trend analysis.
- **Limited fire safety**: Temperature monitoring is reactive, and flame detection or early fire warnings are often absent.
- **No IoT integration**: Without connectivity, users cannot receive alerts or monitor the battery remotely.
- **Poor scalability and adaptability**: They struggle to support advanced EVs with high energy density batteries or additional sensors.
- **Restricted data analysis**: Absence of historical logging prevents early fault detection or long-term performance insights.
- **Rigid protection circuits**: Fixed thresholds may trigger false alarms or delayed responses, reducing accuracy.
- **Minimal user feedback**: Indicators and alarms give limited information, leaving users reactive rather than proactive.
- **Lack of integration**: They often operate in isolation from vehicle subsystems, telematics, or IoT platforms.

Due to these drawbacks, many EVs still rely on outdated systems that cannot fully ensure safety, predictive maintenance, or optimal performance, highlighting the need for IoT-enabled battery monitoring solutions.

DISADVANTAGES

- Limited Real-Time Monitoring: Traditional systems lack IoT connectivity, preventing real-time monitoring, predictive maintenance, and timely identification of battery faults or anomalies.
- No Remote Access Capability: Absence of remote monitoring restricts users from accessing battery status, increasing the likelihood of unnoticed critical issues.
- Inadequate Fire Detection: Existing systems lack efficient flame detection mechanisms, leading to delayed response during battery fires, increasing potential safety hazards.
- Limited Historical Data Analysis: Conventional systems do not store historical data, preventing analysis, fault prediction, and proactive battery maintenance for extended lifespan.
- Passive Protection Mechanism: Systems rely on predefined thresholds, limiting adaptability and resulting in delayed responses or false alarms during abnormal conditions.
- Poor User Interface Feedback: Basic indicators fail to provide detailed insights, preventing users from making informed decisions for battery optimization and safety.

2.2PROPOSED SYSTEM

The proposed IoT-based battery monitoring system enhances the safety, efficiency, and reliability of electric vehicle (EV) batteries through the integration of advanced sensors, automated safety mechanisms, and real-time IoT connectivity. The system continuously monitors crucial parameters of the battery such as voltage, temperature, and fire hazards, while providing both local alerts and remote monitoring features.

Key features of the proposed system include:

- **Voltage Sensor**: Prevents overcharging and deep discharge by continuously monitoring voltage levels and triggering corrective actions.
- **Temperature Sensor**: Detects overheating and thermal runaway risks. If unsafe levels are detected, the Arduino activates the buzzer and disconnects the battery through the relay.

- **Flame Sensor**: Provides early fire detection. On detection, the relay disconnects the battery and the buzzer generates an alert.
- **NodeMCU**: Ensures IoT connectivity, transmitting real-time data to the cloud for monitoring via mobile/web applications. It also supports predictive maintenance and remote fault detection.
- **Arduino UNO**: Serves as the main control unit, processing sensor inputs, applying threshold logic, and executing protective actions.
- **LCD Display**: Displays voltage, temperature, and fire status locally, providing instant user feedback.
- **Relay and Buzzer**: Work together to automatically disconnect faulty connections and issue audible alerts during critical conditions.

The proposed system supports real-time notifications, cloud data storage, and historical trend analysis, which enable predictive maintenance and performance optimization. Its modular and scalable architecture makes it suitable for personal EVs, commercial fleets, and industrial battery applications. In summary, the system ensures continuous monitoring, early hazard detection, automated protection, and remote access, thereby extending battery lifespan and promoting safer electric vehicle adoption.

3.SOFTWARE AND HARDWARE REQUIREMENTS

The software ecosystem forms the backbone of the IoT-based battery monitoring system, enabling seamless communication between hardware components, algorithms, and cloud services. It provides the necessary environment for programming, simulation, testing, and integration of system modules. The ecosystem includes development platforms, libraries, and simulation tools that simplify the design and implementation process. By combining coding environments like Arduino IDE and circuit simulation tools such as Proteus, the software ecosystem ensures efficient development, accurate testing, and smooth operation of both hardware and IoT functionalities.

3.1 SOFTWARE ECOSYSTEM

The software ecosystem ensures smooth interaction between hardware, algorithms, and IoT services. It consists of three main layers: operating system, algorithm libraries, and IoT integration platforms.

3.1.1 ARUDINO

The **Arduino Integrated Development Environment (IDE)** is the main platform for programming Arduino boards, widely used by beginners, students, and professionals for embedded systems and interactive projects.

- **Ease of use**: Uses simplified C/C++, with features like syntax highlighting, auto-completion, and built-in libraries to simplify coding.
- **Board compatibility**: Supports a wide range of Arduino boards, automatically detecting the connected model for flexible hardware use.
- **Code uploading & debugging**: Allows easy compilation and USB transfer of code to boards. Includes a serial monitor for real-time communication, debugging, and data analysis.
- **Community & libraries**: Backed by a large open-source community providing libraries, shields, tutorials, and forums for extended functionality and support.

In summary, Arduino IDE combines an intuitive interface, broad hardware compatibility, and strong community support, making it an essential tool for education, prototyping, and advanced embedded system development.

3.1.2 PROTEUS

Proteus is a widely used Electronic Design Automation (EDA) software for designing, testing, and simulating circuits before physical implementation. Developed by **Labcenter Electronics**, it is popular among engineers, students, and hobbyists.

- **Component library**: Offers a vast range of microcontrollers, sensors, ICs, actuators, and passive parts, with options to create custom components.
- **Accurate simulation**: Models real-world behavior using resistance, capacitance, inductance, and semiconductor properties. Supports both analog and digital simulation for mixed-signal systems.
- **Microcontroller support**: Includes virtual controllers like PIC, AVR, and Arduino. Users can upload firmware and test embedded systems directly in simulation.
- **Debugging tools**: Provides oscilloscopes, logic analyzers, and visualization panels to monitor and troubleshoot circuits in real time.
- **Hardware-in-the-loop (HIL)**: Enables interaction between virtual circuits and physical hardware for advanced testing.

In summary, Proteus accelerates development, lowers prototyping costs, and improves reliability with its extensive library, accurate simulations, and powerful debugging environment, making it an essential tool in electronic design.

3.2 HARDWARE REQUIREMENTS

The IoT-based battery monitoring system is built using essential hardware components that work together to ensure safe and efficient operation. The Arduino UNO acts as the main controller, while the NodeMCU provides IoT connectivity for wireless data transmission. Supporting components like sensors, relays, buzzer, and display units enable effective monitoring, control, and user interaction.

3.2.1 BATTERY CHARGER

The battery charger ensures safe and efficient recharging of the electric vehicle's battery pack. It prevents overcharging and maintains a consistent flow of current during the charging process. In case of power interruptions, it safeguards against data loss and system shutdowns. By ensuring stable and reliable operation, it enhances overall battery life and system efficiency.

3.2.2 POWER SUPPLY CIRCUIT

The power supply circuit provides the necessary regulated voltage and current to all components in the system. It converts AC mains into a stable DC output suitable for sensors and microcontrollers. Both linear and switching supply designs can be used depending on efficiency and load requirements. This circuit ensures uninterrupted operation and protects components from voltage fluctuations.

3.2.3 ARDUINO UNO

The Arduino UNO serves as the main microcontroller responsible for processing input data from sensors and controlling output devices. It coordinates system functions such as monitoring, alerting, and data display. Its compatibility with multiple modules and ease of programming make it suitable for real-time applications. The UNO acts as the decision-making core of the monitoring system.

3.2.4 TEMPERATURE SENSOR (DHT11)

The DHT11 sensor measures both temperature and humidity with reliable accuracy. It sends digital signals directly to the microcontroller, reducing the need for external calibration. Its simple three-pin interface allows quick integration into embedded systems. This sensor ensures the system remains aware of environmental conditions critical to battery safety.

3.2.5 VOLTAGE SENSOR

The voltage sensor module operates based on the voltage divider principle to measure high voltages safely. It reduces the input voltage to a measurable range compatible with the Arduino's analog input pins. This enables accurate monitoring of battery voltage and power levels. The sensor ensures that the battery operates within safe voltage limits.

3.2.6 FIRE SENSOR

The fire sensor detects the presence of smoke, flame, or flammable gases using a semiconductor gas sensing element. It produces analog and digital outputs that are directly read by the microcontroller. When abnormal conditions are detected, it immediately triggers alerts and safety actions. This sensor plays a crucial role in preventing fire-related accidents and ensuring safety.

3.2.7 RELAY

A relay is an electrically operated switch that allows a low-power circuit to control a high-power circuit, providing electrical isolation and safety. It can be interfaced with microcontrollers like Arduino or NodeMCU for automated switching of devices, alarms, or other loads in real-time.

3.2.8 DC MOTOR

The DC motor is used to simulate the load connected to the monitored battery. It demonstrates the system's ability to handle real-time mechanical operations under controlled power. The relay manages the motor's operation during safety tests. This helps in validating the system's response under various load and fault conditions.

3.2.9 LIQUID CRYSTAL DISPLAY

The LCD module displays critical system parameters such as temperature, voltage, and alert status in real-time. It is directly interfaced with the Arduino for continuous updates. The display enhances system usability by providing immediate visual feedback to users. It acts as the primary interface for monitoring system performance

3.2.10 NODE MCU

The NodeMCU module provides Wi-Fi-based connectivity, allowing the system to transmit data to cloud platforms or remote devices. It communicates with the Arduino to receive sensor readings and send them wirelessly. This enables remote supervision of battery performance and early fault detection. The NodeMCU ensures seamless IoT integration for smart monitoring.

3.2.11 BUZZER

The buzzer serves as an audio alert mechanism that activates during abnormal events such as overheating or voltage faults. It helps in instantly notifying the user to take preventive measures.

The buzzer works in coordination with sensors and relay modules to provide timely warnings. Its low power consumption and compact design make it ideal for embedded systems.

3.3 BLOCK DIAGRAM

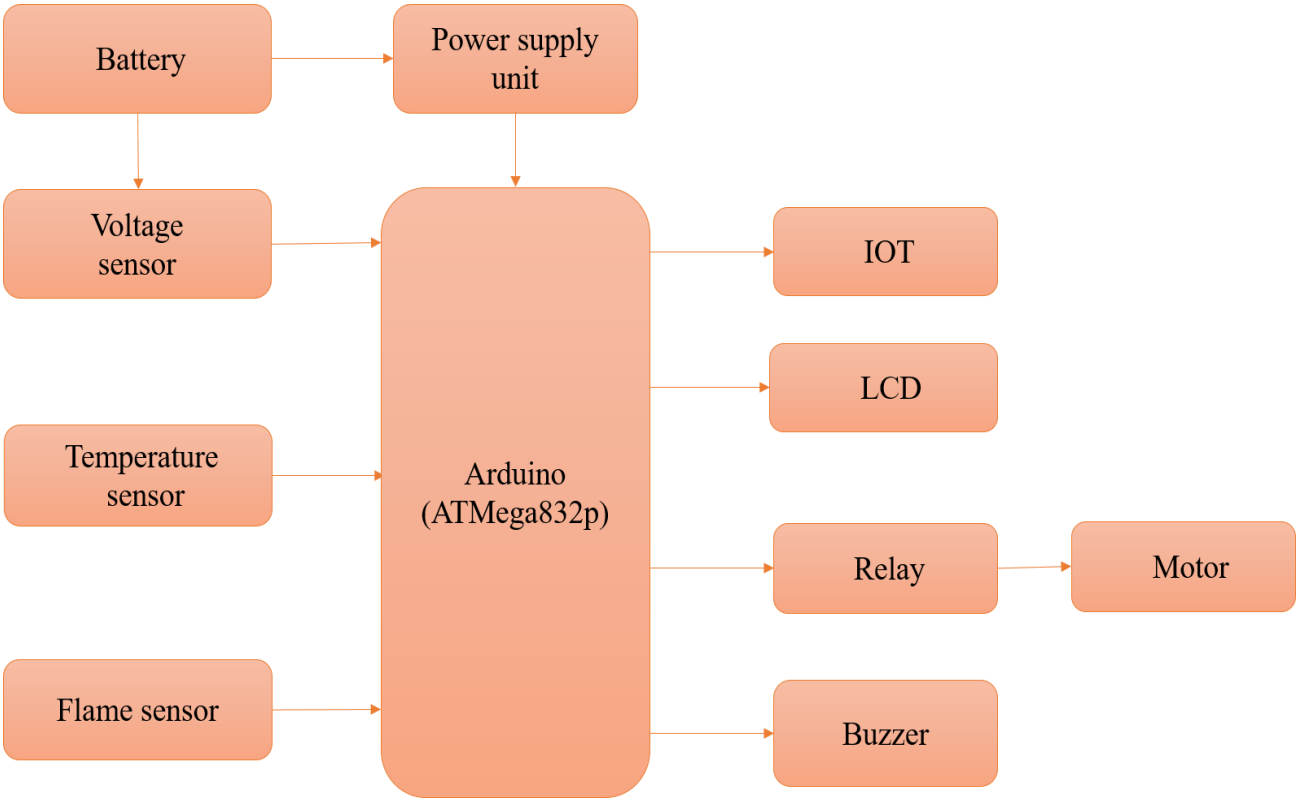

Figure 3.1 – Block Diagram

4.REVIEW OF LITERATURE

4.1 TITLE: Battery Management System with Wireless Parameter Estimation in EV

AUTHOR: Vishnu K., Ojha A., Nema R. K.

YEAR: 2021

DESCRIPTION: This paper introduces a wireless approach to parameter estimation in Battery Management Systems (BMS) for electric vehicles. Traditional BMS often rely on wired sensors, which increases system weight and complexity. The authors propose replacing these with wireless techniques to achieve more flexibility and reduce wiring harness issues. The study demonstrates how wireless data transmission can be applied for real-time monitoring of key parameters like voltage and temperature. Furthermore, the method minimizes maintenance costs and makes the system more scalable. The work highlights how wireless estimation can improve reliability and overall vehicle efficiency in future EV applications.

4.2 TITLE: Smart Battery Management Technology in Electric Vehicle Applications: Analytical and Technical Assessment

AUTHOR: Lipu H., Miah M. S., Ansari S., Wali S. B., Jamal T., Elavarasan R. M., Kumar S., Naushad Ali M. M., Sarker M. R., Aljanad A.

YEAR: 2022

DESCRIPTION: This paper provides a comprehensive review of smart battery management technologies for electric vehicles. It emphasizes analytical and technical methods for enhancing safety, extending battery life, and improving system efficiency. The study highlights how IoT and predictive maintenance techniques can be integrated into EVs. It also discusses challenges such as high energy demand, battery degradation, and the need for accurate state estimation. Through case studies, the authors illustrate the benefits of adopting advanced BMS models. The findings suggest that smart BMS solutions are essential to support large-scale EV adoption and sustainable energy practices in the future.

4.3 TITLE: Electric Vehicle Batteries: Status and Perspectives of Data-Driven Diagnosis and Prognosis

AUTHOR: Zhao J., Burke A. F.

YEAR: 2022

DESCRIPTION: This research paper reviews the role of data-driven techniques in monitoring and predicting battery health for electric vehicles. It explains how machine learning and big data analytics can be applied to diagnose faults and predict failures. The study points out that conventional BMS lack predictive accuracy as batteries age. By incorporating data-driven methods, the system can provide more reliable estimates of battery life and performance. The authors also analyze challenges such as handling large datasets and ensuring real-time accuracy. The study concludes that predictive data-driven approaches will play a key role in advancing EV battery technologies.

4.4 TITLE: Smart Lithium-Ion Battery Monitoring in Electric Vehicles: An AI-Empowered Digital Twin Approach

AUTHOR: Pooyandeh M., Sohn I.

YEAR: 2023

DESCRIPTION: This study explores the use of a digital twin framework, enhanced with artificial intelligence, for monitoring lithium-ion batteries in EVs. A digital twin creates a virtual model of the battery, allowing accurate tracking of its condition in real time. The authors integrate AI to improve SoC and SoH estimation, making fault detection faster and more reliable. This combination enables predictive maintenance by identifying degradation patterns before failure occurs. The proposed system reduces downtime and improves EV performance through proactive interventions. The research demonstrates how digital twin technology, combined with AI, can transform future EV battery management systems.

4.5 TITLE: An IoT-Based System for Monitoring Parameters and Passive Cell Balancing of a Lithium-Ion Battery Pack

AUTHOR: Parthasarathi S., Dhanaselvam J., Saravanakumar K., Ajayan A.

YEAR: 2023

DESCRIPTION: This paper presents an IoT-enabled battery management system designed for real-time monitoring and passive cell balancing. The system uses IoT sensors to collect data such as voltage, current, and temperature, which are then sent to cloud-based dashboards for analysis. The inclusion of passive cell balancing ensures that all cells operate uniformly, extending the battery's lifespan. The study also emphasizes the role of IoT in providing continuous remote monitoring for improved safety. It demonstrates how integrating monitoring with balancing enhances EV performance under varying environmental conditions. Overall, the system ensures both reliability and scalability for lithium-ion packs.

4.6 TITLE: Wireless Battery Management Systems: Innovations, Challenges, and Future Perspectives

AUTHOR: Cao Z., Gao W., Fu Y., Mi C.

YEAR: 2024

DESCRIPTION: This work reviews wireless battery management systems (wBMS) as an innovation in EV design. Traditional wired systems increase complexity and weight, while wireless systems simplify architecture and improve reliability. The authors analyze challenges such as interference, latency, and cybersecurity issues. They also explore how wBMS can improve manufacturing flexibility by reducing wiring costs. The paper discusses innovations in wireless protocols and energy-efficient communication technologies. It concludes by highlighting the future potential of wBMS for large-scale EV deployment, emphasizing their role in making EVs more robust and cost-effective.

4.7 TITLE: Deep Learning Approach for Optimum Power Management Using IoT in EV Battery Management System

AUTHOR: Deepan Kumar S., Arun Raj V., R. V., Manojkumar R.

YEAR: 2024

DESCRIPTION: This research integrates deep learning with IoT for optimizing EV battery power management. The authors propose a neural network model that processes real-time IoT data to balance energy usage efficiently. The system is capable of predicting faults, identifying anomalies, and adjusting energy distribution accordingly. This enhances performance by ensuring stable operation under different driving conditions. The study also emphasizes reduced energy losses and better battery utilization.

5.DEFINITION OF THE PROBLEM AND SCOPE OF THE RESEARCH WORK

5.1 PROBLEM DEFINITION

Electric Vehicles (EVs) are becoming increasingly popular due to their eco-friendliness and efficiency. However, the performance, safety, and lifespan of EVs are highly dependent on the condition of their batteries. Issues such as overcharging, deep discharge, overheating, and fire hazards pose significant risks to battery health and passenger safety.

Traditional battery management systems focus mainly on charge–discharge control but lack advanced monitoring and real-time protective mechanisms. In the absence of immediate detection and corrective actions, abnormal conditions can lead to reduced battery life, costly damage, or even accidents.

Thus, the problem addressed in this research is the lack of an integrated system that can:

- Continuously monitor voltage, temperature, and fire hazards.
- Take immediate protective actions locally (relay cut-off, buzzer alerts).
- Provide IoT-based remote monitoring for predictive maintenance and user awareness.

This project aims to overcome these limitations by designing an IoT-based battery monitoring system that ensures real-time safety, enhances performance, and supports wider adoption of EV technology.

5.2 SCOPE OF RESEARCH WORK

The scope of this research is focused on the design, development, and testing of an IoT-enabled battery monitoring system for electric vehicles. The proposed work covers the following aspects:

Sensor Integration: Implementation of voltage, temperature, and flame sensors to ensure comprehensive monitoring of battery parameters.

Microcontroller-based Control: Utilization of Arduino UNO as the main processing unit for data acquisition, threshold comparison, and safety actions.

IoT Connectivity: Use of NodeMCU (ESP8266) for Wi-Fi connectivity, enabling real-time data transmission to a cloud platform for remote access.

Local Safety Mechanisms: Relay disconnection and buzzer alarms for immediate response during abnormal conditions, ensuring user and vehicle safety.

User Interface: LCD display for on-site monitoring, and mobile/web dashboards for remote supervision.

Data Logging and Analysis: Storage and visualization of historical data for predictive maintenance and performance optimization.

The system is designed to be scalable and modular, making it adaptable for personal EVs, commercial fleets, and industrial battery applications. However, the current scope is limited to prototype development and testing under controlled conditions. Future research may extend this work by integrating artificial intelligence for predictive fault detection, optimizing energy usage, and supporting large-scale deployment in smart transportation networks.

6.ARCHITECTURAL DESIGN

6.1 ARCHITECTURAL DESIGN

The architectural design defines how the system components interact across different functional layers. It ensures scalability, modularity, and reliability.

- **Sensing Layer:**
  Includes the voltage, temperature, and flame sensors responsible for collecting critical battery data.
- **Processing Layer:**
  Arduino UNO acts as the control unit, processing inputs and making decisions. NodeMCU provides IoT connectivity for cloud integration.
- **Actuation & Alert Layer:**
  Relay disconnects the battery in emergencies, buzzer produces audio alerts, and LCD provides instant on-site feedback.
- **Cloud & Data Management Layer**:
  The IoT platform (Blynk) logs sensor data, provides dashboards, and supports historical trend analysis for predictive maintenance.
- **User Layer**:
  Users access battery status and alerts through mobile or web applications, ensuring visibility and remote supervision.

This layered architecture ensures that local safety actions happen instantly, while remote monitoring and analytics provide long-term insights and reliability.

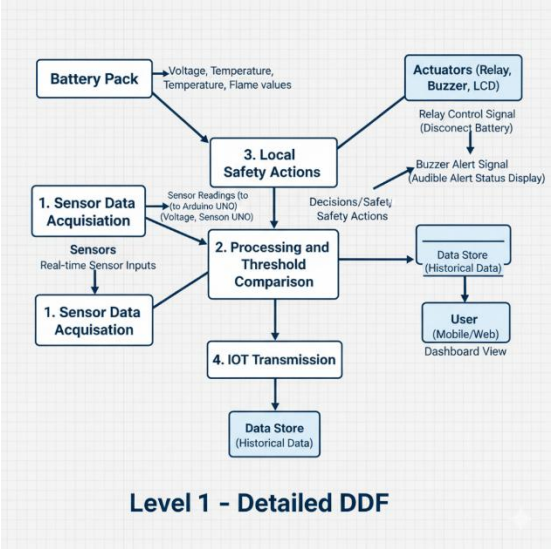

Figure 6.1 – Detailed DFD Diagram

6.2 DATA FLOW DIAGRAMS (DFDs)

Data Flow Diagrams (DFDs) represent how information flows within the proposed IoT-based battery monitoring system. They show how data moves from sensors to processing units, actuators, and the user interface.

Entities and Data Flow:

- Battery Pack: Provides voltage and temperature values to the sensors.
- Sensors: Collect real-time data (voltage, temperature, flame) and pass it to the monitoring system.
- IoT-Based Monitoring System (Process): Processes the sensor data, performs threshold checking, and takes decisions.
- Cloud Server: Receives sensor data via NodeMCU for remote storage and monitoring.
- User: Receives alerts and can view data through a mobile or web dashboard.
- Actuators (Relay, Buzzer, LCD): Provide immediate safety measures and feedback to the user.

Thus, the Context Diagram shows the interaction between inputs (battery and sensors), the system, and outputs (alerts, cloud data, user interface).

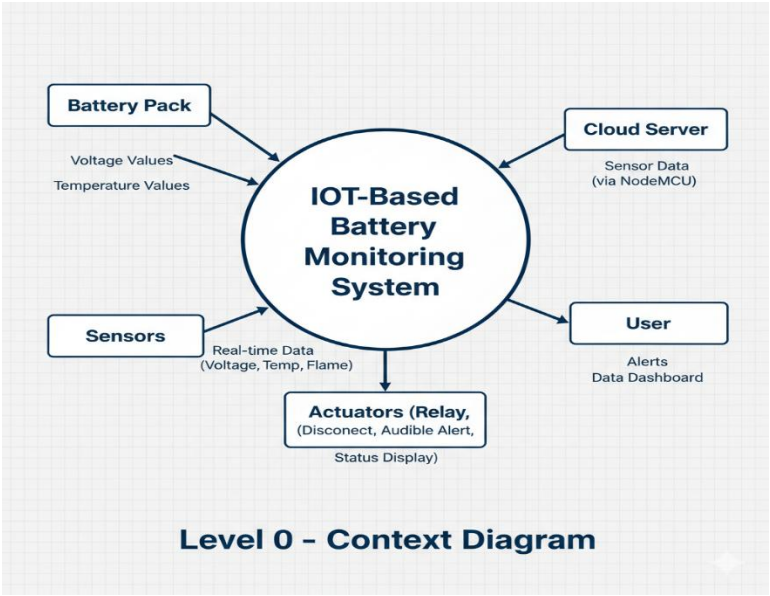

Figure 6.2 – Data flow diagram

7.RESULT AND DISCUSSION

7.1 RESULT

The IoT-based battery monitoring system demonstrated major improvements in **safety, efficiency, and battery lifespan**. With voltage, temperature, and flame sensors integrated into automated safety mechanisms, the system provided real-time monitoring and rapid intervention during abnormal conditions.

- **Safety performance**: During tests, abnormal temperature spikes and flames were detected with high accuracy. The buzzer and relay shutdown responded within milliseconds, preventing thermal runaway, fires, and failures.
- **Real-time monitoring**: NodeMCU enabled cloud-based data transmission, allowing users to remotely track voltage, temperature, and flame status. Local LCD feedback provided on-site updates, creating a dual-layer monitoring system for better situational awareness.
- **Battery health**: Continuous voltage monitoring prevented overcharging and deep discharging, extending battery life. Historical data logging supported predictive maintenance and trend analysis, enabling proactive interventions.
- **Accuracy and reliability**: Sensors performed consistently under stress tests. Flame detection worked even in low-light and noisy conditions, temperature readings were precise across environments, and voltage levels were monitored effectively.
- **Automated protection**: The relay instantly disconnected the battery during hazards, while the buzzer provided clear audible alerts. This minimized human error and enhanced overall safety.

Conclusion of implementation: The IoT-based battery monitoring system proved to be highly reliable and effective in ensuring real-time safety and protection for batteries. By continuously monitoring voltage, temperature, and flame conditions, it significantly reduced the risk of accidents, equipment damage, and operational failures. The system's automated relay and buzzer response minimized human error and provided immediate alerts during hazardous situations. Historical data logging and predictive maintenance features further enhanced battery lifespan and efficiency. Its consistent performance across various conditions demonstrated accuracy and dependability. The dual-layer monitoring, combining local LCD feedback and cloud-based tracking, improved situational awareness for users. Overall, the system not only safeguards battery operations but also builds confidence in the safe deployment of electric vehicles. Its

adoptioncontributes to safer, more efficient, and sustainable energy solutions, making it an contributes to safer, more efficient, and sustainable en solutions, making it an essential essential advancement in modern battery management technology.

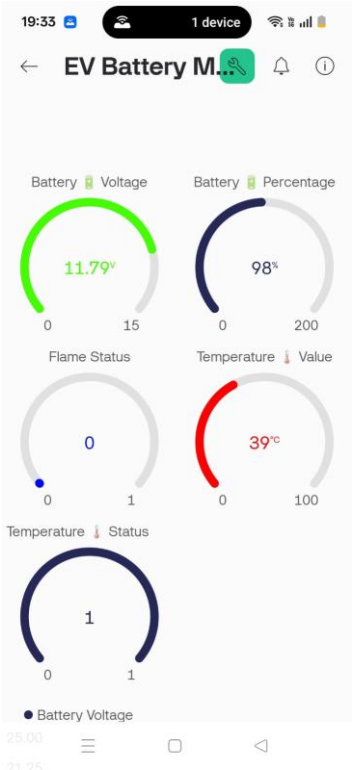

Figure 7.1- Dashboard

7.2 IoT CLOUD APPLICATION

Blynk is an IoT platform that enables real-time monitoring and control of hardware devices through a user-friendly mobile application. In our smart waste management system, Blynk is integrated to display real-time sensor data, trigger alerts, and facilitate remote monitoring. The system leverages the Blynk IoT platform to visualize waste bin levels, fire alerts, toxic gas concentrations, and system status on a cloud-connected dashboard.

Blynk Dashboard Setup

To effectively monitor waste management parameters, a custom Blynk dashboard was designed with the following widgets:

Widgets Used:

- Gauge Widgets: Display real-time waste level percentage.
- LCD Widget: Shows text updates such as “Bin Full” or “Normal.”

- Graph Widget: Plots historical data of gas sensor readings.
- LED Indicator: Turns red for fire detection and green for normal operation.
- Button Widget: Manually resets the alert system.
- Notifications Widget: Sends push alerts when thresholds are exceeded.

Setting Up the Dashboard:

1. Create a new project in Blynk and select the appropriate hardware (ESP8266/ESP32).
2. Add widgets and assign virtual pins to each sensor.
3. Configure notifications for critical alerts like full bin detection, fire hazard, and gas leakage.
4. Link IoT module (ESP8266/ESP32) to the Blynk cloud using the authentication token.
5. Deploy the dashboard and test real-time data updates.

Data Flow and Functionality:

- **Fire Sensor:**
  - Detects flames and triggers an alert.
  - Buzzer activates, and an LED indicator turns red.
  - Notification sent to authorities.
- **Gas Sensor:**
  - Measures toxic gas levels (MQ-135 sensor).
  - When levels exceed safety thresholds, a warning is displayed.
- **Real-Time Alerts:**
  - Alerts are sent via Blynk Notifications and email.
  - Waste management teams receive actionable insights to optimize collection schedules.

Integration with ESP8266/ESP32

The ESP8266/ESP32 module connects the hardware system to the Blynk cloud. Data is transmitted using the Blynk library and an internet connection. The following steps outline the integration:

1. Install Blynk Library in Arduino IDE.
2. Include ESP8266WiFi and BlynkSimpleEsp8266.h libraries.
3. Define Blynk authentication token, WiFi credentials, and virtual pins.
4. Implement sensor data acquisition and upload readings to the Blynk cloud.
5. Use Blynk.run() in the loop function for real-time updates.

Benefits of Blynk Integration:

- Remote Monitoring: Authorities can monitor waste bins in different locations.
- Real-Time Alerts: Quick notifications enhance safety and operational efficiency.
- User-Friendly Interface: No complex coding required, making it accessible.
- Cloud Data Storage: Historical data helps optimize waste collection routes.
- Energy Efficient: Reduces unnecessary waste collection trips, lowering fuel costs.

O/P Screen:

Figure 7.2 - Output Screen

Figure 7.3 - Output Screen

8.CONCLUSION AND FUTURE ENCHANCEMENT

8.1 CONCLUSION

In this paper, we have proposed an IoT-based battery monitoring system for electric vehicles that leverages wireless communication and cloud computing to collect and analyze battery data in real-time. Our system offers granular and accurate insights into battery health and performance, real-time monitoring and analysis capabilities, cloud-based analysis, and enhanced safety. Through our experiments and evaluations, we have demonstrated the effectiveness and reliability of our system in detecting potential issues and providing actionable insights to users. We have also shown that our system can be easily integrated into existing electric vehicle infrastructure and can scale to accommodate large fleets of vehicles. The paper described the design and development of an IoT-based battery monitoring system for electric vehicle to ensure the battery performance degradation. We are developing the system for battery management in electric vehicle by controlling the crucial parameters such as voltage and temperature. It is very important that the BMS should be well maintained with battery reliability and safety. This present paper focusses on the study of Battery Management System and optimizes the power performances of electric vehicles. Moreover, the target of reducing the greenhouse gases can greatly be achieved by using battery management system.

8.2 FUTURE ENHANCEMENT

An IoT-based battery monitoring system in electric vehicles is an essential aspect of ensuring efficient operation and prolonging the battery life of electric vehicles. IoT-based battery monitoring systems can collect a significant amount of data related to battery usage and performance. By applying advanced data analytics, such as machine learning and artificial intelligence, to this data, the system can gain a deeper understanding of battery performance and predict potential issues before they occur. Integrating IoT-based battery monitoring systems with smart grids can enable better control and optimization of the energy flow between the vehicle and the grid. This can help to ensure that the battery is charged efficiently and at the optimal time. Wireless charging technology is becoming more prevalent and can be integrated with IoT-based battery monitoring systems to provide a seamless and effortless charging experience for electric vehicle users. Multi-modal battery management is a system that can manage the performance of

multiple batteries in an electric vehicle. By monitoring and balancing the performance of these batteries, the system can help to optimize battery life and improve overall vehicle performance. Block chain technology can be used to create a secure and transparent record of battery usage and performance. This can help to ensure that batteries are properly maintained and replaced when necessary, reducing the risk of battery failure and improving overall vehicle safety. In the future of IoT-based battery monitoring systems in electric vehicles is exciting, with many potential enhancements that can improve battery performance, vehicle efficiency, and overall user experience. As technology continues to advance, we can expect to see more innovative solutions for battery monitoring and management in electric vehicles.

9.REFERENCES

9.1 REFERENCES

[1] K. Vishnu, A. Ojha, and R. K. Nema, "Battery management system with wireless parameter estimation in EV," in *Advances in Clean Energy Technologies*, Springer Proceedings in Energy, Singapore: Springer, 2021, pp. 639–651. doi: 10.1007/978-981-16-0235-1_47.

[2] H. Lipu, M. S. Miah, S. Ansari, S. B. Wali, T. Jamal, R. M. Elavarasan, S. Kumar, M. M. Naushad Ali, M. R. Sarker, A. Aljanad, *et al.*, "Smart battery management technology in electric vehicle applications: Analytical and technical assessment toward emerging future directions," *Batteries*, vol. 8, no. 11, p. 219, 2022. doi: 10.3390/batteries8110219.

[3] J. Zhao and A. F. Burke, "Electric vehicle batteries: Status and perspectives of data-driven diagnosis and prognosis," *Batteries*, vol. 8, no. 10, p. 142, 2022. doi: 10.3390/batteries8100142.

[4] M. Pooyandeh and I. Sohn, "Smart lithium-ion battery monitoring in electric vehicles: An AI-empowered digital twin approach," *Mathematics*, vol. 11, no. 23, p. 4865, 2023. doi: 10.3390/math11234865.

[5] S. Parthasarathi, J. Dhanaselvam, K. Saravanakumar, A. Ajayan, *et al.*, "An IoT-based system for monitoring parameters and passive cell balancing of a lithium-ion battery pack in a fixed environmental temperature setting," SAE Technical Paper 2023-01-5062, 2023. doi: 10.4271/2023-01-5062.

[6] Z. Cao, W. Gao, Y. Fu, and C. Mi, "Wireless battery management systems: Innovations, challenges, and future perspectives," *Energies*, vol. 17, no. 13, p. 3277, 2024. doi: 10.3390/en17133277.

[7] S. D. Kumar, V. A. Raj, V. R, and R. Manojkumar, "Deep learning approach for optimum power management using IoT in EV battery management system," SAE Technical Paper 2024-28-0085, 2024. doi: 10.4271/2024-28-0085.

[8] V. R, M. SivaramKrishnan, J. N.R, L. K.I, M. P, and S. R, "Intelligent Battery Management System with AI and IoT for Enhanced EV Efficiency," 2025 3rd Int. Conf. on Artificial Intelligence and Machine Learning Applications (AIMLA), Namakkal, India, pp. 1–6, 2025, doi: 10.1109/AIMLA63829.2025.11041561.

[9] S. H. Grandhi, H. M. Al-Jawahry, D. S, B. V. Kumar, and M. K. Padhi, "A Quantum Variational Classifier for Predictive Maintenance and Monitoring of Battery Health in Electric

Vehicles," 2024 Int. Conf. on Intelligent Algorithms for Computational Intelligence Systems (IACIS), Hassan, India, pp. 1–4, 2024, doi: 10.1109/IACIS61494.2024.10721715.

[10] R. X. Goh, H. W. Su, C. M. Cheong, T. F. Lai, M. H. A. Halim, and R. A. Jackson, "AI Based Control System for Electric Vehicles with Predictive Maintenance and Adaptive Energy Management," 2025 21st IEEE Int. Colloquium on Signal Processing & Its Applications (CSPA), Pulau Pinang, Malaysia, pp. 112–117, 2025, doi: 10.1109/CSPA64953.2025.10932986.

10. APPENDIX

Source Code:

```
#include <LiquidCrystal.h>

LiquidCrystal lcd(13,12, 11, 10, 9, 8);


#include <Adafruit_Sensor.h>

#include "DHT.h"

#define DHTPIN 2   // what pin we're connected to

#define DHTTYPE DHT11   // DHT 11


int t;

DHT dht(DHTPIN, DHTTYPE);

//////////////////////////////voltage1////////////////

float correctionfactor1 = 0;

int analogInput1 = A1; //voltage

float vout1 = 0.0;

float vin1 = 0.0;


// two resistors 30K and 7.5k ohm

float R1_1 = 30000;  //

float R2_1 = 7500; //

float value1 = 0.0;

int x,y,z;
```

```
int R;
int Z2,Z1;
int vin1_1;
int s,f;
int F;

void setup()
{
dht.begin();
      Serial.begin(9600);
      lcd.begin (16,2);
      lcd.setCursor(0,0);
      lcd.print("EV Battery ");
      lcd.setCursor(0,1);
      lcd.print("Management System ");
    pinMode(A1,INPUT);  //volt 1
pinMode(2,INPUT);// temp
pinMode(7,INPUT);// fire
pinMode(3,OUTPUT);//motor
pinMode(5,OUTPUT);//buzzer
   delay(3000);
      lcd.clear();
}
```

```
void loop()
{
  dht11();
voltage1();
if( (f==1)||(s==1) )  /////fire and buzzer contion
   {
      digitalWrite(5,HIGH);
        delay(1000);
      digitalWrite(5,LOW);
     }


else if( (f==0)||(s == 0))
{
digitalWrite(5,LOW);
}
F=digitalRead(7);
if(F==LOW)   ////fire sensor
{
  lcd.setCursor(7,1);
  lcd.print("F_H");
f=1;
digitalWrite(3,HIGH);
}
```

```
else
{
lcd.setCursor(7,1);
lcd.print("F_L");
f=0;
digitalWrite(3,LOW);
}

  Serial.println("T");
   Serial.println(t);
   delay(200);
   Serial.println("S");
  Serial.println(s);
  delay(200);
   Serial.println("a");
   Serial.println(vin1);
  delay(200);
     Serial.println("b");
  Serial.println(vin1_1);
  delay(200);

   Serial.println("d");
  Serial.println(f);
```

```
  delay(200);
}
void voltage1()
{
   value1 = analogRead(analogInput1);
   vout1 = (value1 * 5) / 1023.0; // see text
   vin1 = vout1 / (R2_1/(R1_1+R2_1));
   vin1 = vin1+0.7 - correctionfactor1;
   lcd.setCursor(0,0);
   lcd.print("V:");
   lcd.setCursor(2,0);
   lcd.print(vin1);
vin1_1=((vin1/12)*100);
  lcd.setCursor(8,0);
   lcd.print("BP:");
   lcd.print(vin1_1);
   lcd.print("% ");
}
 void dht11()
{
int h = dht.readHumidity();
        t = dht.readTemperature();
       float f = dht.readTemperature(true);
```

```
        if (isnan(h) || isnan(t) || isnan(f))
        {
          //Serial.println("Failed to read from DHT sensor!");
          return;
        }
         lcd.setCursor(11,1);
         lcd.print("T:");
         lcd.setCursor(13,1);
         lcd.print(t);
 if(t > 38)   /////temp
        {
     lcd.setCursor(15,1);
 lcd.print("H");
s=1;  ///temp high motor offf buzzer on
// digitalWrite (4,HIGH);
        }
        else
        {
      lcd.setCursor(15,1);
      lcd.print("L");
  s=0; /////temp low motor on buzzer offf
//    digitalWrite (4,LOW);
        }}
```